BUSINESS DATA MANAGEMENT

Lecture05

Demand Curve and Supply Curve

Hello friends. In the last video, we tried to understand some of the fundamental concepts of demand and supply. In this video, we will go a step further and understand what is called the demand curve, what is called the supply curve, what drives the shift in demand curve and supply curve, and how does price get determined, which is called equilibrium pricing. So let's get started. So as we discussed last time, a demand curve shows the relationship between the quantity demanded versus the price.

And as you can see in the graph before you, As we see that there is a shift in price of a particular product, you can see there is a shift in the quantity that is demanded. So the demand of a particular good or service typically would increase when the price points come down. Okay.

For example, in this particular case that we discussed even in the last video, when the price point is one, the quantity demanded for this particular good is eight. When the price point doubles to two, the quantity demanded moves from eight to four. So the Key point here is that the demand curve is downward sloping when you look at it in the context of price point which is the y-axis and the quantity demanded which is on the x-axis. So as you can see when there is a demand that is increasing, the price point for the same goes on reducing.

So the price point at $3, at the price point of $3 here, the quantity demanded is nothing, which is zero. However, when the price point comes down to as low as half, then the quantity demanded moves to 10 units of this particular case. So what is a shift in the demand curve? What drives the shift in demand? So the demand curve shifts when any variable that affects buyer's desire to purchase alters the quantity demanded at any given price point.

We all know that an increase in demand refers to the change that raises the quantity demanded at a price point. The shift in demand curves to the right. Any decrease in demand reduces the quantity that is demanded at that particular price point and therefore the shift to the curve goes to the left. So if there is an increase in demand the shift will move this particular blue line will move from left to right and when there is a decrease in demand this particular blue line would move from right to left.

So what derives what what are the variables that shift this demand curve okay now there are a number of variables that shift the demand curve but some of the key variables that you as a data analyst you as a business analyst needs to focus or needs to analyze are the following ,number one you should look at the income levels of your consumers if the income level increases typically demand for normal good also increases. For example, if there is the employment salary levels increase, you will see that the demand for consumer products, consumer goods, such as let's say autos, they also rise. If the demand rises when the income falls, it essentially means that the good that we're talking about is actually not a normal good, it's an inferior good. So when your income rises, when a consumer's income rises, or the consumer's income falls, they will start taking the bus instead of buying a car, that particular

Phenomena means that this particular good is an inferior good. So it's important to understand what product the business is dealing in and what will be the impact of the change in the income levels. Now when you're analyzing the shift in demand for your particular product, it is also important to analyze how the price points are behaving for related products or related goods which could either be substitutes or it could be complementary goods. Let's take an example.

For example, let's say there are two goods that are substitutes. You can either eat an ice cream or you can have a frozen yogurt. If there is a fall in the price of one, it reduces the demand for the other, so if there's a fall in price of ice cream, it reduces the demand for frozen yogurt. These goods are called substitute goods, okay? Because there's a fall in price, we know when we started the

demand curve, there is going to be a rise in demand for that particular good. When two goods are complementary goods, if there's a fall in the price of one, it increases the demand for the other.

For example, if you can either eat an ice cream or let's say a hot fudge, and they're usually consumed together, if there's a fall in the price of one, the demand for the other also increases because you have these two goods together. They are consumed in complementarity. Complementarily. The third and a very important variable that shifts the demand curve, either increases the demand or reduces the demand, are the preferences of taste of your market, of your consumers. For example, a fitness trend, which is a change in taste or some kind of a trend that you can see, that will have an effect on the increase in demand for related goods,

with related services, for example, protein shakes. So we'll see there's a surge in demand for protein shakes also because there is a shift in mindset of consumers to move towards more fitness. Future expectations, the expectations of how the price point will be in the future, might also impact demand. For example, And this is very much in the context of today's geopolitical situation.

If the consumers expect that the fuel price hike will happen, they might want to increase their petrol purchases as of now. so that they can circumvent the rise in prices in the future. So that shift in demand will again happen because there has been a perception that there could be some kind of a change in the future. Another example that can again lead to a shift in demand is when there is an increase in the buyer segment. A good example will be, let's say, if you look at a particular location and that particular location, let's say a company is setting up their factory and because of them setting up a factory, there is going to be an employment of, let's say, 2,000

you know, people who will be now employed, they will be blue-collared, white-collared employees in that particular factory. Now you will see that suddenly there will be a shift in demand for consumer products and other kinds of

durable products, et cetera, because there has been an increase in the buyer segment. So the key point here is that Yes, there is a strong relationship between price and demand, but any shift in demand can happen due to many variables, some of these variables we just discussed. When you are looking as a data analyst into the demand expectations of your particular product, when you're trying to forecast, it's important to look at it in the context of how the other variables, both in the short term and long term, are expected to behave.

Okay, now let's take a specific case study around cigarette smoking, and let's say we are looking at how there can be a shift in demand for cigarettes. In this particular case, we will discuss how there will be a downward shift in demand; there will be a reduction in the demand for cigarettes. So let's say the public policymakers want to reduce the amount of people who are into cigarette smoking, and because of that, they apply a measure of announcing publicly, advertising the bad effects of cigarette smoking. They go all out educating the consumers about how cigarette smoking is injurious to their health, and we have seen that that has happened in the last few decades. Now there's no change in price point per se, but you will see that there has been a shift in demand. So if you look at this particular graph on the right-hand side, the original demand at a particular price point was 20, 20 units. At that particular price point, you can see point number A. You can see the intersection at A.

But because of this whole discouragement through public announcements by government and policymakers, there's been a shift in demand curve from A to B. Mind you, the price point remains the same, but the number of cigarettes being consumed goes down. So that's an example, one example of how there can be a shift in demand through the mechanism of public announcements and advertisement. The other way that can discourage potential consumers to smoke would be if there has been a shift in price point. So let's say the government wants to discourage smoking, and now they start applying some additional taxes that the suppliers are now forced to change their price point.

Because we know that when there's a change in the input prices of the products, the suppliers become less encouraged to sell the same quantity at the same price. So what they will typically do is they will increase the price point. In this case, if you see, because the government has levied certain taxes, the price point moves from two to four. And as you can see, there is no shift in the demand curve, but there is a change in the quantity that is demanded, that moves down from 20 to 12. We call this the movement along the demand curve.

So one is what we learned was the shift in the demand curve. There's a structural shift in the demand curve. The other is there's a movement in the demand, the same demand curve because of a change in price point. In this particular case, it moves from two to four. OK, so let's look at the same concept of curves.

in the context of supply. So as we discussed in the last video, we see that the supply curve is upward sloping, which means that with every increase in price point, the supplier's propensity would be to increase the quantity that they want to supply. They want to sell because they want to make more profit. That's the underlying motivation. Now, there can be shift in supply curves as well.

What drives this shift in supply curve? The way we try to understand from a consumer perspective, it is also important to look at it from a supplier's perspective. As we just discussed in the previous example, any change in the input prices of the good that is being sold, or any change in the prices which will lead to the services being sold, If there is an increase in that price point, now for the producers or the suppliers, this particular good becomes less profitable because the input costs move up, whereas your selling price remains the same. So your profits get squeezed.

So that's one variable which leads to a shift in the supply curve. The other reason which is happening as we speak is the advancements of in technology and which typically would reduce the labor costs. So once the labor costs get reduced, there's more mechanization, your ability to have more economies of scale would increase. And therefore, there will be a positive shift in the supply because now

your cost of production, your cost of making that particular good would probably go down because of these advancements in technology, because of process enhancements.

and so on. So technology would typically lead to an increase in the supply of goods because of economies of scale, because of better efficiency and effectiveness. The third is the expectation. If the firm expects the market price to rise in the future, it might want to store some of the current production with the expectation of making that available in the market in the future.

So in the interim period, there would be a reduction in the supply, but that supply would be made available at a higher price point in the future. Again, if you can see the link here that this is another determinant or factor or variable that can lead to a shift in the supply curve. As we discussed in the demand curve analysis, where we said, hey, the number of buyers will increase the demand. If the number of sellers change, that also has a shift in the supply curve because the capacity to that extent goes up or down. For example, if there has been a number of sellers who have left that particular business,

So the capacity to produce and provide and sell that good goes down to that extent. So again, number of sellers is going to be another factor that can be attributed to a shift in the supply curve. Mind you, all these factors and variables that we're talking about, they are not exhaustive. You can think about other variables as well in the context of your product or the service for which you are analyzing, for which you're doing your data analysis.

Let's look at bringing demand and supply constructs and concepts together. And it's important to understand that the price point gets determined when there is an intersection of demand and supply. And this price point is called the equilibrium price. Equilibrium is the state when the price has reached the level where the quantity supplied meets the quantity demanded, where both quantity supplied and quantity demanded are equal. Okay, so this is also the price point that we say balances the quantity supplied and quantity demanded.

Okay, now what happens when this whole equilibrium quantity which is either supply or demand changes? For example, when do you have a surplus situation? A surplus situation is if the market price is above the equilibrium price. What I mean by that is because the market price is above the equilibrium price, the suppliers would be motivated to provide more of those goods. But the consumers do not want to buy those goods at that market price because it is higher than the equilibrium price.

So what will be the situation? This situation will lead to an excess supply situation or what is also called as the surplus situation. And the response to this typically the businesses would do would be by cutting prices by bringing down the prices so that they are able to allure and they are able to market their product at the equilibrium price which pushes more sales and this again leads to going back the price point which is above the equilibrium price down back to the equilibrium price.

So the price that was earlier made available to the consumers was at a higher than the equilibrium price and because of this shift in the seller's mindset of bringing the price down, they are able to bring back the quantity supply and the quantity demanded together. And at that particular price, they go back to the equilibrium price. The reverse of that is a situation when the market price is below the equilibrium price. So what happens when the market price is below the equilibrium price?

There is more demand. When there's more demand, the supplies are not able to keep up with those demands. And mind you, when the price is lower than the equilibrium price, the suppliers are also not motivated to sell or produce the goods at the lower price. So what happens in this situation? In this situation, the sellers respond by raising the price back to the equilibrium price, okay, so that they are able to meet the demand, the supply and the demand expectations.

So it's important to understand that in a market, in a free market scenario, the price of any good ultimately naturally adjusts to bring the quantity supplied and

the quantity demanded into a balanced state. And that is really the law of demand and supply. The price for any good based on the market scenario would naturally adjust to bring back the quantity supplied and the quantity demanded into a state of balance. Now let's look at this a little bit more graphically to better understand the point that we just discussed about equilibrium. If you see this particular graph on the right hand side, you will see that there is this whole demand curve.

And then there you have, if you superimpose the supply curve on here on the same graph, you will see that from a consumer's perspective, lower the price, higher the demand. We discussed that, right? That's the demand curve. From a buyer's, from a producer's or a seller's perspective, higher the price, more the supply. Now when a point in which when both of these meet, when the demand curve and the supply curve meet, is called as the equilibrium point.

From a quantity perspective, it is the equilibrium quantity, the balance that we were talking about, and from a pricing perspective, it will be the equilibrium price. Okay, so that's how equilibrium price and equilibrium quantity gets determined through this intersection of demand and supply. Now let's try and analyze and this is something which again you will do as data analyst. You will try and have these what if scenarios of what happens when there is a shift in either supply or shift in demand or shift in both to the price point.

Ultimately, it's important to understand that you are trying to determine what is the best price that your firm, your business can sell your product to be able to get maximum returns or profits. So if you look at this particular graph on the right hand side you see the focus on the blue lines you have d1 which is your demand curve and you have the supply curve at this point in time you are able to sell seven quantities at a price point of two okay now suppose in this particular case you have a situation, a variable that there is this is summer season because if this particular product is ice cream at the summer season what happens is that there is a there is a need there is an underlying demand for higher quantity of ice

creams to be consumed okay so what happens here is that your demand curve shifts from D1 to D2 Mind you, your supply remains the same.

There's no change in the supply. So what happens here is that when your demand moves from D1 to D2, there's a new equilibrium price that gets constructed here, gets identified here. So the price moves from two to two and a half with this shift in demand given the weather condition, which is one of the variables. Okay, so again, when you look at things as a data analyst from a seasonal perspective of your particular good, you might want to then shift your pricing based on the expectation of the underlying demand. So an event that raises quantity demanded at any given price shifts the demand from D1 to D2 here.

The equilibrium price and the equilibrium quantity, both in such situations, they increase as we can see from the graph. Okay, here the driver is the season, the abnormal summer season that enables or directs the buyers to demand more of this particular product, which is ice cream. So the demand curve shifts from D1 to D2, which shifts the equilibrium price as we discussed from two to two and a half, and the equilibrium quantity moves from seven to 10. So this is in a nutshell, the understanding of what we mean by demand curves, supply curves, the shift in the demand curves and shift supply curve, the intersection of demand curve and supply curve and the determination of price and how price can change based on the changes in variables, both demand side variables and supply side variables.

I hope you learned from this particular video, and I look forward to meeting you in the next video. Thank you and bye for now.